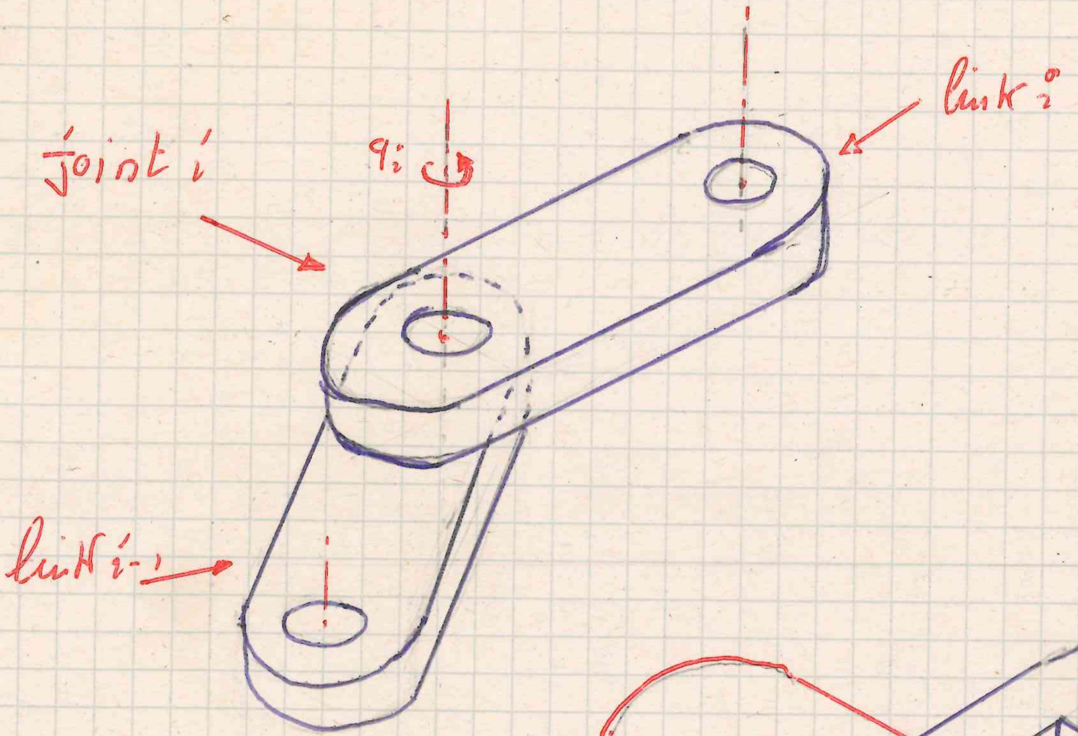
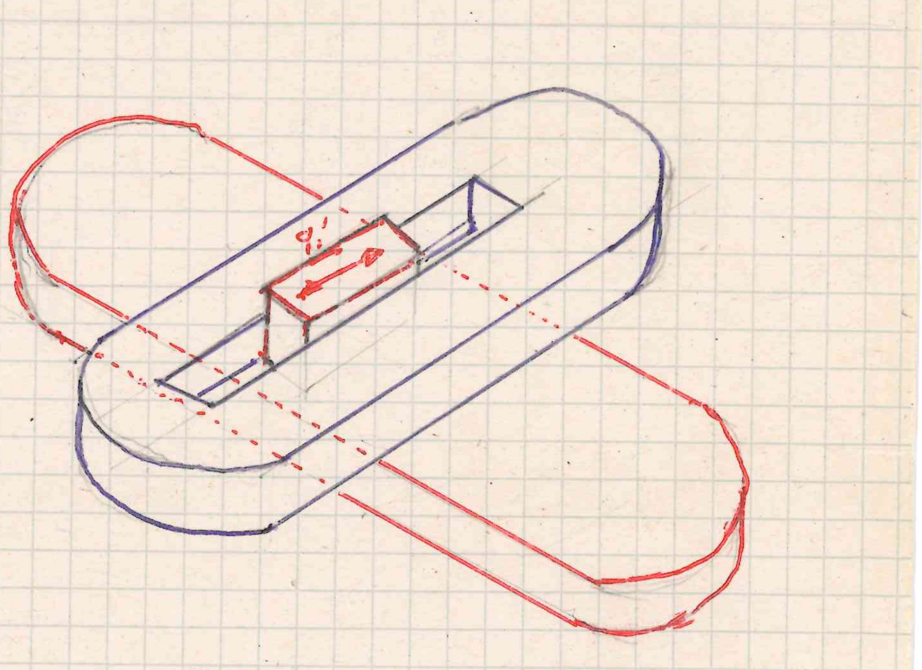
DYNAMICS OF KINEMATIC CHAINS

KINEMATIC PAIRS specify relative motion constraints between two rigid bodies. In robotics the most common are specified by rotational joints (RJ) and by translational joints (TJ) (*)(**)





(*) Rotoidal and prismatic pairs.

(**) Spherical joints are also sometimes used but not treated here.

Kinematic pairs specify <u>constraints</u>

A constraint is any "entity" limiting the motion of a body (or a set of bodies)

Given N bodies any function of the type

$$f(\underline{x}_1 \cdots \underline{x}_N, {}^{0}_{1}R, \cdots, {}^{0}_{N}R, t) \geq 0 \text{ specifies } \begin{cases} = 0 & \text{bilateral positional constraints} \\ \geq 0 & \text{unilateral positional constraints} \end{cases}$$

$$f(\underline{x}_1 \cdots \underline{x}_N, \underline{v}_1 \cdots \underline{v}_N, {}^{0}_{1}R \cdots {}^{0}_{N}R, \underline{\omega}_{1/0} \cdots \underline{\omega}_{N/0}, t) \geq 0 \text{ specifies } \begin{cases} = 0 & \text{bilateral kinetic constraints} \\ \geq 0 & \text{unilateral kinetic constraints} \end{cases}$$

If f depends explicitly on t the constraint is <u>rehonomic</u>
If f does not depend explicitly on t " " " <u>scleronomic</u>

Bilateral positional constraints are also called <u>holonomic constraints</u>

Kinematic pairs are <u>holonomic constraints</u>.

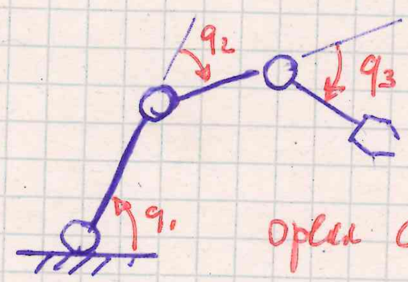
A robot formed by <u>holonomic joints</u> is said holonomic robot

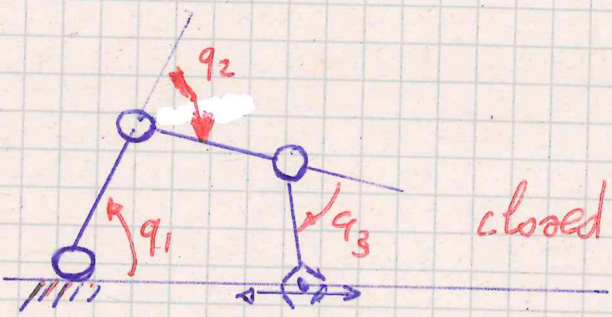
All other types of constraints are called <u>non-holonomic</u>

For the case of kinematic pairs it is natural to associate to each constraint a parameter (joint variable) which specify the relative position/orientation of the bodies forming the pair.

A sequence of bodies constrained by kinematic pairs defines a kinematic chain (*)(**)

In open kinematic chains the joint variables can be arbitrarily specified (***). In closed kinematic chains the joint variables are further constrained





For open chains two smooth (continuous and differentiable) functions of $q = (q_1 \cdots q_n)^T$ can be defined for each body.

nr. of kinematic pairs

***) Joints limits always exist, but are neglected in this discussion.

(**) Kinematic chains can be possibly branched (e.g. robot hand)

*) One of the bodies is usually connected to ground. If this is not the case the kinematic chain is floating (e.g. flying/floating robots)

$$\begin{cases} \underline{x}_{P_i/0} = \underline{x}_{P_i/0}(\underline{q}) \\ {}^{0}_{i}R = {}^{0}_{i}R(\underline{q}) \end{cases}$$

These functions define $\forall \underline{q}$ the configuration of the bodies forming the kinematic chain (*)

The analytical expression of these functions is in general complex but they can be computed in recursive form.

NOTE The functions above are continuous then for "small" variation of q the position and orientation of the bodies forming the chain is expect to change by a "small" amount.

In particular for each body

$$d\underline{x}_{P_i/0} = \frac{\partial \underline{x}_{P_i/0}}{\partial \underline{q}} d\underline{q}$$

$$d\,{}^{0}_{i}R = \left[\frac{\partial\, {}^{0}_{i}R_{ij}}{\partial \underline{q}} d\underline{q} \right]$$

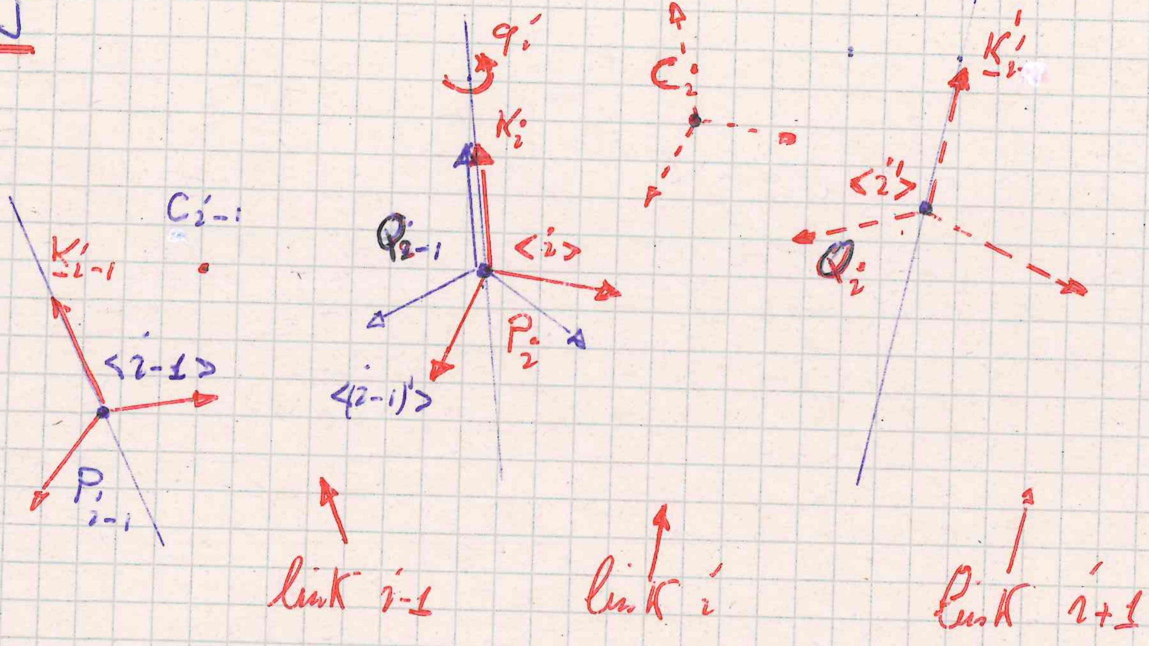
$[d\underline{\theta}_i \times]\,{}^{0}_{i}R$

as of Poisson formula

these partial derivatives are directly related to the Jacobian matrices.

(*) P_i generic point on body $\mathcal{B}_i$ (a.k.a. link)

RJ



$$P_i \equiv Q_{i-1} \qquad \left(P_i - P_{i-1}\right) \triangleq \underline{r}_{i/i-1}$$

constant in frame <i-1>

$${}^{i-1}_{\;\;i}R \equiv \underbrace{R_{i-1}}_{\text{constant}} \cdot R_z(q_i)$$

rotation about z axis

$$\underline{\omega}_{i/i-1} = \underline{K}_i \, \dot{q}_i$$

$$\begin{cases} \underline{v}_{i/0} = \underline{v}_{i-1/0} + \underline{\omega}_{i-1/0} \times \underline{r}_{i/i-1} \\ \underline{\omega}_{i/0} = \underline{\omega}_{i-1/0} + \omega_{i/i-1} = \underline{\omega}_{i-1/0} + \underline{K}_i \, \dot{q}_i \end{cases}$$

$$\begin{cases} \underline{\dot{v}}_{i/0} = \underline{\dot{v}}_{i-1/0} + \underline{\dot{\omega}}_{i-1/0} \times \underline{r}_{i/i-1} + \underline{\omega}_{i-1/0} \times \left(\underline{\omega}_{i-1/0} \times \underline{r}_{i/i-1}\right) \\ \underline{\dot{\omega}}_{i/0} = \underline{\dot{\omega}}_{i-1/0} + \underline{\dot{K}}_i \, \dot{q}_i + \underline{K}_i \, \ddot{q}_i = \underline{\dot{\omega}}_{i-1/0} + \underline{\omega}_{i-1/0} \times K_i \dot{q} + \\ \qquad + \underline{K}_i \, \ddot{q}_i \end{cases}$$

$$\underline{\dot{K}}_i = \underline{\omega}_{i/0} \times \underline{K}_i = \underline{\omega}_{i-1/0} \times \underline{K}_i + \underbrace{\underline{K}_i \, \dot{q}_i \times \underline{K}_i}_{\underline{0}}$$

$$\begin{cases} \underline{v}_{c_i/0} = \underline{v}_{i/0} + \underline{\omega}_{i/0} \times \underline{r}_{c_i/i} \\ \underline{\omega}_{c_i/0} = \underline{\omega}_{i/0} \end{cases}$$

$$\begin{cases} \underline{\dot{v}}_{c_i/o} = \underline{\dot{v}}_{i/o} + \underline{\dot{\omega}}_{i/o} \times \underline{r}_{c_i/i} + \underline{\omega}_{i/o} \times (\underline{\omega}_{i/o} \times \underline{r}_{c_i/i}) \\ \underline{\dot{\omega}}_{c_i/o} = \underline{\dot{\omega}}_{i/o} \end{cases}$$

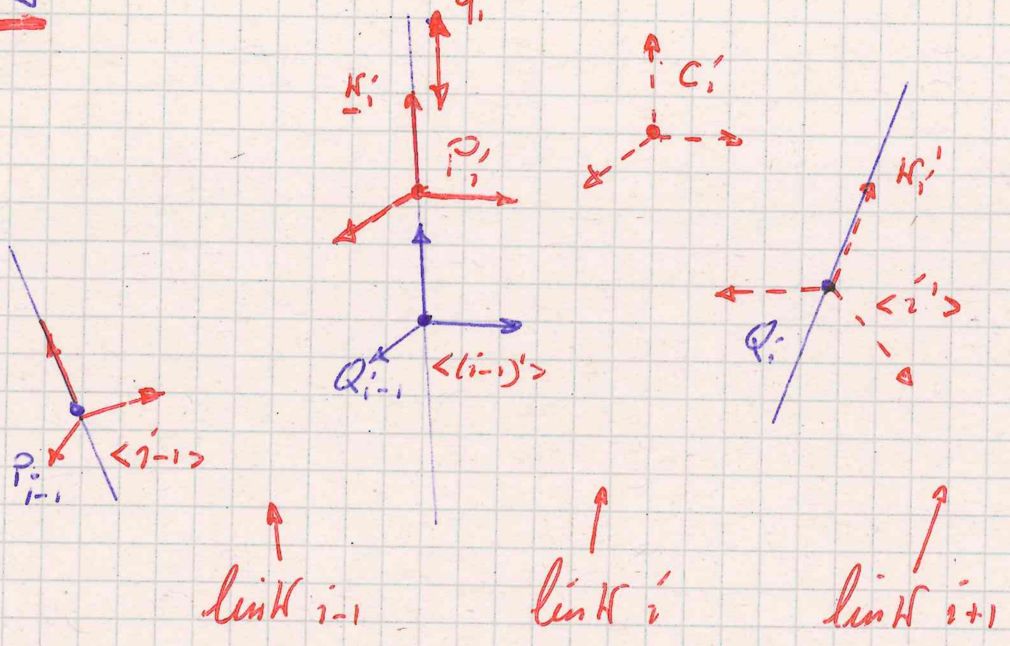
NOTE The choice of P_i, Q_i and of the matrix R_i may follow procedures like Denavitt-Hartemberg algorithm.

The baricentric ref. frame is chosen to be coincident to <i> for sake of simplicity additional (local) rotations could be specified if required.

NOTE The above formulas for the computation of velocity and acceleration are recursive and apply from the ref. frame <o> up to the reference point of interest (typically the end effector)

NOTE For homogeneity of notation when needed the point O (origin of the inertial frame <o>) is denoted as P_0. The choice of P_0 is arbitrary.

IV



$$\underline{r}_{i/(i-1)'} = (P_i - Q_{i-1}) = \underline{k}_i \, q_i$$

$$^{i-1}_{\ i}R = R_{i-1} \quad \text{constant}$$

$$\underline{\omega}_{i/i-1} = \underline{0}$$

$$\begin{cases} \underline{v}_{i/0} = \underline{v}_{i-1/0} + \dfrac{d_{i-1}}{dt} \underline{r}_{i/i-1} + \underline{\omega}_{i-1/0} \times \underline{r}_{i/i-1} = \\ \underline{\omega}_{i/0} = \underline{\omega}_{i-1/0} \end{cases}$$

$$\underbrace{\dfrac{d_{i-1}}{dt} \underline{r}_{i/i-1}}_{\underline{k}_i \dot{q}_i}$$

$$= \underline{v}_{i-1/0} + \underline{\omega}_{i-1/0} \times \underline{r}_{i/i-1} + \underline{k}_i \dot{q}_i$$

$$\begin{cases} \dot{\underline{v}}_{i/0} = \dot{\underline{v}}_{i-1/0} + \dot{\underline{\omega}}_{i-1/0} \times \underline{r}_{i/i-1} + \underline{\omega}_{i-1/0} \times (\underline{\omega}_{i-1/0} \times \underline{r}_{i/i-1} + \underline{k}_i \dot{q}_i) \\ \qquad + (\underline{\omega}_{i-1/0} \times \underline{k}_i) \dot{q}_i + \underline{k}_i \ddot{q}_i \\ \dot{\underline{\omega}}_{i/0} = \dot{\underline{\omega}}_{i-1/0} \end{cases}$$

NOTE $\underline{v}_{c_i/0}$, $\underline{\omega}_{c_i/0}$, $\dot{\underline{v}}_{c_i/0}$, $\dot{\underline{\omega}}_{c_i/0}$ are computed as for RV –